

From Tool Access to Research Capability: Tools and Memory for Financial Agents

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Abstract

Language models can invoke external quantitative libraries, yet raw tool access and unconstrained verbal self-reflection (**Reflexion**) fail to produce sound financial research decisions when domain conventions are fragmented, execution outputs contain look-ahead leaks, or iterative prompt mutations overfit non-stationary regimes via reward hacking. We present **FIN-SKILLS**, an open ecosystem of 129 source-verified financial skills, 48 algorithm backends, and 37 executable verification guards integrated within a closed-loop agent architecture that couples two-stage point-in-time progressive skill routing (**RAG-JEV**), a 64-Kenyon-Cell mushroom-body associative memory module, and **Governed Financial Recursive Self-Improvement (Fin-RSI)**. Across $M = 5$ seeds on **FINGUARDBENCH-60** and a 207,742-row two-dimensional company $\times$ year balanced panel (2018–2026, distilled from 4.23M cross-market records across 5,122 A-share and 81 US tickers), progressive routing compresses prompt context by 98.9% ($4,039 \pm 42$ tokens) with 0.0% schema fragmentation and zero future leaks, while counterfactual guard feedback lifts verified repair pass rates from $17.5\% \pm 1.8\%$ to $96.2\% \pm 1.1\%$. Coupling verified episode returns with gated mushroom-body plasticity improves annualized Sharpe from -0.58 ± 0.11 to $+1.71 \pm 0.14$ across 21 regimes, while evolving multimodal feature and covariance operators inside our SHA-256-locked **Fin-RSI** sandbox lifts sequence effective sample size by $3.62\times$, raises out-of-sample Daily Rank IC from $+0.0283 \pm 0.0038$ to $+0.0363 \pm 0.0042$, and achieves a $+1.77 \pm 0.34$ annualized net Sharpe (DSR = 1.000, Max DD = -6.84%) with Pareto dominance across sparse small-cap and crisis slices.

1 Introduction

A financial research agent must decide which method suits a task, what data that method requires, how to interpret its output, and when later evidence warrants revising its decision. For example, a forecast, a volatility estimate, and a portfolio allocation address different questions. An agent can execute all three correctly yet fail to connect them to the overall objective. Tool access expands the available actions; research capability concerns how those actions are chosen and applied.

This distinction becomes critical across repeated decisions and iterative self-improvement. Useful domain conventions and parameter schemas can be severed by fixed-window RAG chunking, while unconstrained verbal self-reflection (**Reflexion**) in non-stationary financial markets is highly vulnerable to back-test overfitting and reward hacking: when an agent freely modifies both its signal logic and evaluation scripts, it rapidly exploits look-ahead leaks, concentrates weights into mega-cap outliers, or collapses effective sample size during event bursts. Progressive skill routing, verified cross-episode associative memory, and physically governed operator evolution address complementary dimensions of this challenge. Empirical feedback must be cryptographically isolated from the agent’s mutable workspace so that synaptic and algorithmic updates reflect verified, defect-free out-of-sample evidence.

We investigate these questions through **FIN-SKILLS**, an open library linking dated financial guidance with discoverable, executable quantitative methods, and **Governed Financial Recursive Self-Improvement (Fin-RSI)**. An agent can inspect input requirements, select parameters, execute an algorithm, and evolve feature-pooling and covariance-adaptation operators inside a strictly isolated sandbox. We connect this interface with

three core mechanisms: two-stage progressive skill routing with strict point-in-time (`as_of`) gating (`RAG-Jev`), a persistent 64-Kenyon-Cell mushroom-body associative memory gated by settlement-cleared portfolio returns, and a SHA-256-locked `Fin-RSI` evolution loop driven by three physical diagnostic probes (Pre-LayerNorm Scale-Cancellation, Intraday Burst Effective Sample Size ESS, and Subspace Signal-to-Noise Heterogeneity) under a three-way Pareto promotion gate.

Our autonomous-use and recursive evolution studies provide a clear empirical test: while open-weight models successfully executed 43 quantitative algorithms across 64 portfolio decisions, unguided tool execution and unconstrained verbal reflection failed to improve net risk-adjusted returns over unassisted baselines. Without executable counterfactual guards, point-in-time memory gating, and physical operator diagnostics, models frequently selected unsuited algorithms, suffered severe drawdown degradation under regime shifts, or overfitted sparse small-cap slices.

We examine four concrete questions in our empirical investigation: how autonomous tool access alters the quantitative methods an agent selects and its resulting portfolio risk; how atomic two-stage progressive skill routing with point-in-time gating compares against fixed-window chunk `RAG`; whether verified cross-episode returns can guide subsequent method selection under shifting market regimes; and how governed recursive operator evolution (`Fin-RSI`) prevents reward hacking while improving macro, tail-slice, and crisis-regime performance. In summary, this paper makes four primary contributions:

- **Standardized Quantitative Skill & Guard Library (FIN-SKILLS):** We release 129 source-verified financial skills, 48 unified algorithm backends, and 37 executable research guards (including `check_panel_balance` and `check_trial_ledger`) with cryptographic AST-and-data receipt binding.
- **Closed-Loop Routing & Verified Associative Memory:** We formulate a two-timescale architecture pairing point-in-time two-stage progressive skill routing (`RAG-Jev`) with a 64-Kenyon-Cell mushroom-body associative memory gated exclusively by settlement-

cleared, guard-verified episode log-wealth returns.

- **Governed Financial Recursive Self-Improvement (Fin-RSI):** We introduce a physically grounded operator evolution framework that locks the read-only evaluation harness via SHA-256 cryptographic hashes and evolves multimodal pooling, per-channel positive-part James–Stein shrinkage (James and Stein, 1961), and streaming rank-1 Sherman–Morrison–Woodbury inverse-covariance adaptation (Sherman and Morrison, 1950) coupled with Bernoulli Fisher information gating under a three-way Pareto promotion criterion.
- **Rigorous Multi-Seed Empirical & 2D Panel Benchmarks:** Across $M = 5$ independent random seeds, 64 autonomous portfolio decisions, 21 cross-board market episodes, and a 207,742-row company $\times$ year balanced panel (2018–2026, constructed from 4,233,499 multi-source records across 5,122 A-share and 81 US tickers), we isolate the causal contributions of counterfactual guard repair, two-dimensional panel conditioning ($\Delta\text{Rank IC} = +0.0734$, $\Delta\text{Sharpe} = +3.17$, $t = 20.14$, $p = 1.2 \times 10^{-11}$), JEV System-One isotonic calibration (Sharpe = +1.986), associative plasticity, and four-generation `Fin-RSI` evolution (Rank IC = $+0.0363 \pm 0.0042$, Sharpe = $+1.77 \pm 0.34$, ESS ratio 3.62 $\times$).

2 Related work

Financial agent frameworks and benchmarks.

Classical quantitative portfolio rules such as equal-weight diversification ($1/N$) (DeMiguel et al., 2009), time-series momentum (TSMOM) (Moskowitz et al., 2012), and hierarchical risk parity (HRP) (López de Prado, 2016) provide transparent, leak-free baselines but cannot synthesize unstructured disclosures or adapt across shifting tool catalogs. Recent LLM agent frameworks augment language models with reasoning loops, episodic memory, and multi-agent debate: ReAct (Yao et al., 2023) and Reflexion (Shinn et al., 2023) interleave tool execution with verbal self-critique; FinMem (Yu et al., 2024) introduces layered short-, mid-, and long-term FAISS episodic memory; TradingAgents (Xiao et al., 2024) or-

chestrates bull–bear analyst debates; and FinAgent (Zhang et al., 2024) combines multimodal market signals with tool reflection. Meanwhile, InvestorBench (Kou et al., 2024), StockBench (Chen et al., 2025), and the Finance Agent Benchmark (Bigeard et al., 2025) evaluate sequential financial decisions. However, existing frameworks rely on natural-language reflection or dense chunk retrieval without executable counterfactual verification or locked evaluation harnesses, leaving them vulnerable to silent look-ahead joins, unadjusted corporate actions, and iterative backtest overfitting. Our study isolates the effects of tool access, point-in-time progressive routing, counterfactual guards, associative memory, and governed recursive operator evolution under identical market panels and transaction costs.

Verification guards, cross-episode plasticity, and governed self-improvement. Behavioral testing offers a structured view of model capabilities beyond aggregate scores (Ribeiro et al., 2020). Financial backtest overfitting and selection bias (Bailey and López de Prado, 2014; Bailey et al., 2017), alongside look-ahead leakage in financial agents (Li et al., 2025), motivate explicit checks on decision evidence. Our persistent memory component adapts the fruit-fly mushroom-body model of Huang et al. (2024) to update sparse associative readouts exclusively from guard-verified, settlement-cleared financial returns, while Fin-RSI complements empirical-Bayes shrinkage (James and Stein, 1961; Ledoit and Wolf, 2004) and rank-1 matrix updates (Sherman and Morrison, 1950) with cryptographic harness isolation and multi-slice Pareto governance.

3 Framework: tools, routing, guards, memory, and governed evolution

We define research capability as the ability to choose appropriate quantitative methods, interpret their outputs, and adapt both future choices and underlying feature operators based on realized outcomes without look-ahead leakage. Counting API calls alone does not measure this capability. Our framework integrates progressive skill routing, counterfactual execution guards, cross-episode associative memory, and governed financial recursive self-improvement (Fin-RSI).

Decision, memory, and governed evolution cycle. Let q_t be the task prompt, $\mathcal{D}_{\leq t}$ the mar-

ket data available at time t , and M_t the persistent memory state. A two-stage progressive router with point-in-time (`as_of`) filtering selects atomic skill cards that fit the context budget without chunk fragmentation. The agent uses the hydrated skill specification and a readout of M_t to select a tool, execute it with chosen parameters, and produce an asset allocation. When the trade clears at settlement $t + T$, the realized return updates M_t , provided it passes all 37 verification checks. Across multi-generation research cycles, Fin-RSI diagnoses mathematical bottlenecks in the feature and covariance operators and mutates candidate operators within a SHA-256-locked evaluation sandbox under a three-way Pareto promotion criterion.

Complementary retrieval, memory, and operator evolution timescales. Progressive routing operates within a single task by hydrating only the 1–2 complete `SKILL.md` packages relevant to the query under a strict `as_of` timestamp filter. Persistent memory operates across trading episodes by storing sparse Kenyon-Cell readout updates in an external module, while Fin-RSI operates across generational research cycles by evolving the mathematical form of multimodal feature pooling, subspace shrinkage, and streaming covariance adaptation inside an immutable evaluation harness.

Grounded feedback and anti-reward-hacking governance. When an episode concludes, the agent records the market context, the chosen tool, and the realized log-wealth return. Only verified, settlement-cleared outcomes update memory. Negative returns are retained as learning signals, while trades relying on look-ahead data, unbalanced panels, or unadjusted prices are rejected by guards to prevent memory corruption. Similarly, during Fin-RSI operator evolution, any candidate mutation that alters the read-only evaluation harness, triggers a counterfactual guard violation, or regresses on sparse small-cap or crisis slices is automatically rejected.

4 Professional tools and domain knowledge

FIN-SKILLS exposes financial methods alongside the information an agent needs to run them: task scopes, input requirements, dependencies, parameter conventions, and dated source verifications.

Component	Role in research workflow	Target question	Implementation and validation
FIN-SKILLS	Document conventions and execute quantitative algorithms	Which methods does the model select, and how are outputs applied?	129 skills, 48 backends; evaluated on 64 portfolio decisions
Progressive routing (RAG-Jev)	Hydrate atomic skill cards under strict point-in-time (as_of) cutoffs	How to avoid chunk fragmentation and future-document leakage?	$4,039 \pm 42$ tokens (-98.91%), 100% recall, 0.0% frag, 0 leaks
Persistent memory	Update associative weights from completed episode returns	Does past experience improve future method selection across regimes?	64-KC mushroom-body model; $+1.71 \pm 0.14$ Sharpe across 21 episodes
Execution checks	Bind verification to code artifacts and filter invalid feedback	Are execution outputs, panels, and learning signals free of leaks?	37 guards (incl. <code>check_panel_balance</code>); 207,742-row 2D panel
Governed evolution (Fin-RSI)	Evolve pooling, shrinkage, and covariance operators in a locked sandbox	Can an agent self-improve operators without backtest overfitting?	SHA-256 lock + 3 probes; $+1.77 \pm 0.34$ Sharpe (ESS $3.62\times$)

Table 1: System components, roles, and validation scope within the closed-loop financial agent framework.

Domain skills route requests to relevant libraries; per-library skills provide implementation guidance. The algorithm catalog covers portfolio construction, forecasting, volatility estimation, technical signals, and execution schedules, providing uniform programmatic contracts over quantitative methods.

Agent-selected execution. An agent can discover candidate algorithms, inspect their requirements, and execute them with custom parameters. Structured outputs separate numerical results from missing dependencies or runtime errors. An optional strategy adviser provides rule-based suggestions without overriding model choices. In our evaluation, agents access 18 price-compatible algorithms, with requests and executions logged independently from free-form text.

Information and model interfaces. Retrieval interfaces chunk packaged skills and documents, returning relevant passages under strict as-of cutoffs to prevent look-ahead bias. Forecasting and prediction adapters provide a unified fit-predict lifecycle, while portfolio optimizers expose specialized execution interfaces. Appendix B details the complete interface inventory, model lifecycle, and retrieval pipeline.

5 Cross-episode memory and governed operator evolution

Foundation language models are stateless across sessions: once an interaction ends, the model retains no memory of whether its previous trading strategy made or lost money. Fine-tuning the entire LLM with gradient descent after each trading

day is impractical, computationally expensive, and prone to overfitting on noisy financial markets.

To provide cross-episode adaptation without altering LLM weights, we couple an external associative memory module inspired by the fruit-fly mushroom body (Huang et al., 2024) with **Governed Financial Recursive Self-Improvement (Fin-RSI)**. In biological systems, the mushroom body maps complex sensory features into sparse representations and uses dopamine-gated synaptic plasticity to associate sensory cues with delayed rewards and punishments across multiple decay rates. Complementarily, `Fin-RSI` evolves the mathematical form of the multimodal pooling, subspace shrinkage, and covariance-adaptation operators across generational research cycles inside a SHA-256-locked evaluation sandbox.

Associative memory and complete-episode updates. Our memory module projects market state features and candidate actions into a high-dimensional sparse representation (64 Kenyon Cells). When the agent evaluates a new trading decision, it queries the memory module for associative preference scores based on prior experiences in similar market regimes. Because financial rewards are delayed over a holding horizon T , for positive account values $W_0, \dots, W_T$, the target episode return is

$$G = \sum_{t=0}^{T-1} \log(W_{t+1}/W_t) = \log(W_T/W_0). \quad (1)$$

This return accounts for transaction fees, daily price fluctuations, and final liquidation. The verification interface checks timestamp ordering, return accounting, and data lineage; trades relying

on look-ahead data or unadjusted prices are rejected before reaching the memory module, preventing spurious profits from corrupting associative weights. Valid negative returns remain active learning signals, and each verified trade receipt updates memory exactly once after settlement clears.

Governed Financial RSI (Fin-RSI) physical operator evolution. While associative memory adapts readout weights within a fixed feature parameterization, raw multimodal financial sequences suffer from three structural pathologies diagnosed by our physical probes: (i) pre-LayerNorm scalar weight cancellation ($\|\nabla_{w_t} \text{LN}(w_t \mathbf{h}_t)\|_2 = 0$), (ii) effective sample size ($\text{ESS} = (\sum_t w_t)^2 / \sum_t w_t^2$) collapse during intraday announcement or forum bursts, and (iii) small-cap subspace variance inflation under uniform scalar shrinkage. Guided by these probes inside a read-only evaluation harness (HARNESS_LOCK.json), Fin-RSI progressively evolves the multimodal operator across three mathematical generations:

1. **Gen-1 (Post-Encoder Value-Space Bounded-Saliency & Burst Conservation):** To eliminate pre-LayerNorm scale cancellation and bound single-post leverage during intraday bursts, Fin-RSI moves saliency weighting into post-encoder value space with temperature-bounded activation ($\tau = 2.5$) and Log-Sum-Exp (LSE) intraday burst partition conservation:

$$w_t = \exp(\tau \tanh(z_t/\tau)), \quad \bar{\mathbf{v}}_{i,t} = \frac{\sum_{m \in \mathcal{B}_{i,t}} w_m \mathbf{v}_{i,t,m}}{\sum_{m \in \mathcal{B}_{i,t}} w_m}, \quad (2)$$

lifting sequence effective sample size by $3.62\times$ ($\text{ESS} : 4.61 \rightarrow 14.72$).

2. **Gen-2 (Per-Channel Subspace Precision Gate & Positive-Part James–Stein Denoising):** Because fundamental catalyst channels and noisy retail chatter exhibit heterogeneous signal-to-noise ratios across $D = 16$ latent dimensions, Fin-RSI replaces scalar shrinkage with a per-channel precision gate $\alpha_{i,t,d}$ conditioned on local effective sample size $n_{i,t}^{\text{eff}}$ and cross-modal divergence $\Delta_{i,t,d} = |\mathbf{q}_{i,t,d}^{\text{soc}} - \mathbf{q}_{i,t,d}^{\text{ann}}|$, paired with positive-part James–Stein empirical-Bayes shrinkage

(James and Stein, 1961):

$$\alpha_{i,t,d} = \frac{n_{i,t}^{\text{eff}}}{n_{i,t}^{\text{eff}} + \kappa_d(1 + \Delta_{i,t,d})}, \quad \hat{\theta}_{i,t,d}^{\text{JS}+} = \mu_{0,d} + \left(1 - \frac{(D-2)\sigma_d^2}{\|\bar{\mathbf{q}}_{i,t} - \mu_0\|_2^2}\right)_+ (\tilde{q}_{i,t,d} - \mu_{0,d}). \quad (3)$$

3. **Gen-3 Champion (Streaming Rank-1 Woodbury Covariance Adaptation & Fisher Gate):** To capture non-stationary cross-channel correlations without $\mathcal{O}(D^3)$ batch inversion or look-ahead leakage, Fin-RSI maintains a streaming rank-1 Sherman–Morrison–Woodbury inverse covariance matrix (Sherman and Morrison, 1950) with exponential forgetting factor $\gamma \in (0, 1)$ and modulates the 64-KC readout via a Bernoulli Fisher information uncertainty gate $g_{\text{Fisher}}(p_t) = 4p_t(1 - p_t)$ on the isotonic-calibrated probability $p_t \in (0, 1)$:

$$\Sigma_t^{-1} = \gamma^{-1} \Sigma_{t-1}^{-1} - \frac{\gamma^{-2} \Sigma_{t-1}^{-1} \mathbf{u}_t \mathbf{u}_t^\top \Sigma_{t-1}^{-1}}{1 + \gamma^{-1} \mathbf{u}_t^\top \Sigma_{t-1}^{-1} \mathbf{u}_t}, \quad g_{\text{Fisher}}(p_t) = 4p_t(1 - p_t), \quad (4)$$

where $\mathbf{u}_t = \sqrt{1 - \gamma} \hat{\theta}_t^{\text{JS}+}$. A candidate generation is promoted only if it simultaneously passes all 37 counterfactual guards and dominates predecessors across macro returns, sparse small-cap slices, and crisis regimes.

Isolating learning from risk filtering. To verify that performance gains reflect genuine associative learning rather than passive risk reduction, our benchmark compares the updating memory against a frozen-memory control and a volatility exposure gate under identical historical splits and costs (FLY_GATE_CONTROL_RESULTS.json). On a single daily BTC-USDC trajectory (720 bars, 5 seeds), plastic fly achieves +5.58% mean test return vs. +2.39% for frozen ($\Delta = +3.19\%$) and +3.26% for linear ordinary without the volatility gate, whereas under the static 80th-percentile volatility gate both fly_gated (−0.58%) and frozen_gated (−0.02%) are constrained on the single crypto trend path. Across 21 independent market trajectories (7 equity boards $\times$ 3 non-overlapping 3-year regimes, 2018–2026, 207,742 2D-balanced rows, $M = 5$ seeds) with complete-episode checked feedback ($G = \log(W_T/W_0)$), holding

the 80th-percentile volatility gate, transaction fees (8 bps), and chronological splits strictly identical, `learn_with_gate` achieves a mean annualized Sharpe of $+1.71 \pm 0.14$ compared to -0.58 ± 0.11 for `frozen_with_gate` ($\Delta = +2.29 \pm 0.15$, paired $t = 8.36$, $p < 10^{-7}$, winning 21/21 trajectories), $+0.84 \pm 0.16$ for un-gated `learn_no_gate`, and $+0.17 \pm 0.09$ for `dense linear ordinary_with_gate` (which cannot separate non-linear contrarian-vs.-momentum regime interactions that sparse 64-KC Kenyon Cell expansion resolves).

6 Execution and feedback verification

Tool execution and memory updates require verifiable records of what actually ran. The execution controller computes cryptographic hashes over input data, executed code, and returned allocations. If an agent modifies an artifact after execution, previous validation receipts become invalid. The controller returns `PASS` if all required guards pass, `FAIL` if a guard detects a defect, and `INCOMPLETE` if evidence is missing. This mechanism enforces execution provenance without modifying the underlying algorithm.

For cross-episode memory, timing and accounting checks verify that delayed outcomes cleared settlement before they update persistent state. Trades that rely on look-ahead data or unadjusted corporate splits are blocked, preventing corrupted feedback from distorting the agent’s memory. Unsettled trades wait for settlement, while valid unprofitable trades remain eligible learning signals.

The audit study evaluates models under unconstrained generation, optional self-checking, and mandatory verification. Appendix A describes the evaluation setup, defect detection benchmarks, and empirical audit results across model tiers.

7 Evaluating use, retention, and adaptation

We evaluate agent research capability across three dimensions:

Method selection and task quality. Tasks present distinct quantitative problems, requiring the agent to select tools, set parameters, and interpret numerical outputs. We evaluate whether chosen algorithms match input data properties (e.g., stationarity, regime type, asset universe). Measuring method choice alongside returns separates sound reasoning from fortunate price paths.

Matched controls isolate the contributions of raw price tables, text guidance, fixed tool outputs, and autonomous execution.

Context retention and progressive routing.

Holding the base model, input data, and tool interfaces fixed, we compare full-library concatenation, fixed-window BM25 chunk retrieval, and two-stage progressive skill routing (RAG-Jev) under strict point-in-time (`as_of`) filtering. We measure portfolio decision quality, retention of atomic parameter schemas and executable guard signatures, prompt token footprint, and code-vs.-guidance fragmentation. This determines whether two-stage manifest routing eliminates chunk fragmentation without degrading reasoning accuracy.

Experience and adaptation. Holding tool access and context budgets fixed, we evaluate sequential adaptation by comparing unaugmented models, frozen-memory baselines, and our multi-timescale memory module. All agents receive identical cleared returns. We track how tool choices shift following losses, measuring turnover costs and performance across regime changes. Factorial comparisons test interactions between progressive skill routing and memory updates.

Evaluation metrics include forecast accuracy, portfolio Sharpe ratio, turnover, and token cost. All model outputs and portfolio weights are saved and frozen prior to scoring.

8 Empirical study: tool access and decision outcomes

We evaluate how autonomous tool access affects portfolio decisions in open-weight language models under sequential multi-asset tasks.

8.1 Experimental setup

We test Qwen2.5-Coder 7B and 14B under four conditions: historical prices alone; prices with static skill excerpts; prices with precomputed tool outputs; and autonomous access to the library’s algorithm catalog. Each condition runs across inference seeds 11 and 23, with a budget of five interaction turns per decision (capped at 256 generated tokens per response). Conditions run in randomized order.

In the autonomous condition, the agent receives descriptions of available tools. It can inspect algorithm specifications, run an algorithm with custom arguments, or query a rule-based strategy adviser. Tool calls are dispatched against trailing

Condition	7B	14B
No library	+0.100	−3.324
Skill text	−2.930	−4.567
Fixed tool outputs	−3.075	−3.030
Autonomous library	−3.984	−3.783

Table 2: Mean cumulative proxy return (%) across inference seeds with 5 bps transaction costs per side.

market snapshots and return structured numerical results. In our evaluation, agents access 18 price-compatible quantitative routines.

A decision without valid portfolio weights leaves existing allocations unchanged. Requests, outputs, and intermediate reasoning traces are logged separately, and final weights are frozen before scoring.

8.2 Data and evaluation protocol

We evaluate on daily exchange rates for EUR against USD, JPY, GBP, and CHF from ECB reference data. Messages use anonymized asset identifiers and relative indices rather than calendar dates.

Each decision observes 126 trailing days. The evaluation spans the first 84 trading days of 2025 (ending May 2), with four rebalancing points spaced 21 days apart. Decisions at step t execute at day $t + 1$ prices to prevent look-ahead bias. Portfolios are long-only without leverage, with unallocated capital held in cash. We charge 5 basis points per side for rebalancing and final liquidation. Cash, equal-weight buy-and-hold, and deterministic rules serve as reference baselines.

8.3 Observed execution and returns

Across 16 trials, models generated 64 final portfolio allocations. Autonomous 7B made 13 successful tool calls, while autonomous 14B made 30. Both models recovered from minor JSON formatting errors within their turn budget, completing all decisions without runtime crashes.

Tool choices diverged across model scales: 7B selected cross-sectional momentum, Bollinger reversion, and volatility-targeted momentum, whereas 14B predominantly used Bollinger reversion alongside trend-following indicators.

Table 2 shows returns across all arms. While autonomous 14B improved over its text-only condition by 0.783 percentage points, neither autonomous model beat the unassisted baseline (+0.100% for 7B, −3.324% for 14B).

Analysis of portfolio holdings reveals that tool access drove aggressive risk-taking: mean risky asset exposure for 7B rose from 0.500 without the library to 0.969 with autonomous tools; for 14B, exposure shifted from 1.000 (all-in) to 0.500. Autonomous runs used 38 model calls per model across eight decisions, totaling 1,155,782 prompt tokens and 10,165 completion tokens.

These results illustrate the difference between tool execution and research capability: open-weight models can discover and run quantitative algorithms, but unassisted models struggle to match methods to data regimes or manage portfolio risk.

8.4 Comparison with external financial agent baselines

To situate our closed-loop architecture against established paradigms, we benchmark nine methods across three baseline families over the 21 multi-market trajectories (207,742 2D-balanced rows distilled from our 4,233,499-record, 1.25 GB cross-market longitudinal corpus spanning 2,272,556 EastMoney Guba posts across 464 tickers, 1,544,478 A-share announcements across 5,122 tickers, 281,981 Xueqiu longitudinal posts across 1,004 tickers, and 134,484 US StockTwits messages across 81 tickers; 2018–2026) and FINGUARBENCH-60 ($M = 5$ seeds, Table 3): (1) *Classical quantitative rules*: Equal-Weight ($1/N$) (DeMiguel et al., 2009), 12-month volatility-scaled Time-Series Momentum (TSMOM) (Moskowitz et al., 2012), and Hierarchical Risk Parity (HRP) (López de Prado, 2016); (2) *Single-agent tool and memory frameworks*: unguarded ReAct tool execution (Yao et al., 2023), Reflexion verbal self-critique (Shinn et al., 2023), and FinMem layered FAISS episodic memory (Yu et al., 2024); and (3) *Multi-agent debate and multimodal reflection*: TradingAgents bull–bear debate (Xiao et al., 2024) and FinAgent multimodal tool reflection (Zhang et al., 2024).

As shown in Table 3, verbal reflection (Reflexion) and dense episodic memory (FinMem) reduce defect rates compared to unguarded ReAct but still permit 29.4%–38.5% silent look-ahead and split-adjustment leaks because natural-language self-critique cannot inspect runtime perturbation invariance. Multi-agent bull–bear debate (TradingAgents) consumes $3.8\times$ more tokens (15,420 vs. 4,039) while achieving lower out-of-sample Sharpe ($+0.58 \pm 0.15$) than Fin-

Method / Baseline	Paradigm	Pass@1 (%)	Leak Rate (%)	Prompt Tokens	Daily Rank IC	Net Sharpe ($M = 5$)	Max DD
Equal-Weight ($1/N$) (DeMiguel et al., 2009)	Classical Quant	—	0.0%	0	$+0.0000 \pm 0.0000$	$+0.21 \pm 0.04$	-34.2%
TSMOM (Moskowitz et al., 2012)	Classical Quant	—	0.0%	0	$+0.0064 \pm 0.0011$	$+0.38 \pm 0.06$	-24.8%
Hierarchical Risk Parity (López de Prado, 2016)	Classical Quant	—	0.0%	0	$+0.0089 \pm 0.0010$	$+0.52 \pm 0.05$	-16.4%
ReAct + Tools (Yao et al., 2023)	Unguarded Tool Agent	17.5 ± 1.8	64.2%	13,535	-0.0142 ± 0.0021	-0.31 ± 0.12	-38.6%
Reflexion (Shinn et al., 2023)	Verbal Self-Critique	42.8 ± 2.3	38.5%	11,280	$+0.0048 \pm 0.0017$	$+0.29 \pm 0.11$	-27.5%
FinMem (Yu et al., 2024)	Layered FAISS Memory	51.4 ± 2.1	29.4%	9,840	$+0.0112 \pm 0.0016$	$+0.62 \pm 0.13$	-21.9%
TradingAgents (Xiao et al., 2024)	Bull-Bear Debate	56.2 ± 2.5	26.7%	15,420	$+0.0105 \pm 0.0019$	$+0.58 \pm 0.15$	-23.4%
FinAgent (Zhang et al., 2024)	Multimodal Reflection	61.0 ± 2.2	21.8%	12,650	$+0.0146 \pm 0.0015$	$+0.79 \pm 0.14$	-18.2%
FIN-SKILLS Closed-Loop (Gen-0)	RAG-Jev + 37 Guards + 64-KC	96.2 ± 1.1	0.0%	4,039	$+0.0292 \pm 0.0014$	$+1.71 \pm 0.14$	-10.82%
FIN-SKILLS + Fin-RSI Champion	+ Woodbury-Fisher RSI (Gen-3)	96.2 ± 1.1	0.0%	4,039	$+0.0363 \pm 0.0042$	$+1.77 \pm 0.34$	-6.84%

Table 3: Comparison against classical quantitative baselines and state-of-the-art LLM financial agent frameworks across FINGUARBENCH-60 and 21 multi-market episodes (207,742 2D-balanced rows from 4.23M cross-market records, $M = 5$ seeds).

Mem ($+0.62 \pm 0.13$). By combining atomic two-stage skill routing (RAG-Jev), 37 counterfactual verification guards, and checked 64-KC mushroom-body plasticity, our static closed-loop baseline achieves $96.2\% \pm 1.1\%$ Pass@1, 0.0% leakage, and a $+1.71 \pm 0.14$ net Sharpe ($t = 11.84, p < 10^{-5}$ vs. FinAgent), while evolving the multimodal pooling, subspace shrinkage, and streaming Woodbury-Fisher covariance operators inside our SHA-256-locked Fin-RSI sandbox further lifts out-of-sample Daily Rank IC to $+0.0363 \pm 0.0042$ and net Sharpe to $+1.77 \pm 0.34$ (Max DD = -6.84%).

8.5 Systematic component-wise ablation and governed Fin-RSI evolution

To quantify the causal contribution of each architectural pillar, Table 4 reports a seven-arm leave-one-out ablation starting from the FIN-SKILLS closed-loop architecture:

Every ablated subsystem causes a statistically significant drop in out-of-sample performance: replacing progressive routing with fixed-window BM25 chunking severs prose rules from guard signatures (25.0% fragmentation, $\Delta\text{Sharpe} = -0.57$); removing executable counterfactual guards allows illusory look-ahead returns to poison synaptic updates ($\Delta\text{Sharpe} = -1.63$); freezing associative plasticity under the identical volatility gate causes the largest Sharpe collapse ($\Delta\text{Sharpe} = -2.29, p < 10^{-7}$); and removing two-dimensional company $\times$ year panel balancing exposes the model to mega-cap recency bias ($\Delta\text{Sharpe} = -2.13$ in closed-loop episodes, and $\Delta\text{Rank IC} = +0.0734$, $\Delta\text{Sharpe} = +3.17$, $t = 20.14, p = 1.2 \times 10^{-11}$ in standalone 4-condition panel evaluation; Appendix A). Complementarily, coupling JEV System-One isotonic probability calibration (ECE : $0.0408 \rightarrow 0.0096, -76.4\%$) with an 80th-percentile volatility gate lifts 5-year all-weather OOS Sharpe

from $+1.392$ (uncalibrated hard-threshold gate) to $+1.986$ (CAGR = $+12.44\%$, Max DD = -3.14% , Calmar = 3.96), while Barra-orthogonal multimodal catalyst fusion (MMAN) raises 5-day Rank IC from $+0.0131$ (sparse social posts) to $+0.0508$ ($t = 11.50, p < 10^{-29}$).

Finally, evaluating four generations of operator self-improvement across $M = 5$ disjoint seeds on the 207,742-row 2D-balanced panel (107,999 return observations; full ledger in Appendix A.6, Table 9) demonstrates why financial self-improvement must be governed by cryptographic harness locks and physical diagnostic probes. Whereas unconstrained verbal self-reflection (Unconstrained Verbal RSI) degrades out-of-sample Daily Rank IC to $+0.0045 \pm 0.0024$ and Net Sharpe to $+0.18 \pm 0.23$ (Max DD = -22.47%) due to iterative overfitting on burst outliers, our governed Fin-RSI trajectory monotonically improves every Pareto dimension from the static multimodal baseline (Rank IC = $+0.0283 \pm 0.0038$, Sharpe = $+1.16 \pm 0.22$): Gen-1 restores post-encoder gradient flow and lifts burst sequence ESS by $3.62\times$ (Rank IC = $+0.0287 \pm 0.0041$, Sharpe = $+1.30 \pm 0.13$); Gen-2 eliminates small-cap variance inflation via per-channel positive-part James-Stein denoising, lifting sparse small-cap/STAR slice Rank IC from $+0.0236 \pm 0.0069$ to $+0.0290 \pm 0.0075$ (Sharpe = $+1.51 \pm 0.28$); and the promoted Gen-3 Champion couples streaming rank-1 Sherman-Morrison-Woodbury covariance adaptation with Bernoulli Fisher information gating to reach $+0.0363 \pm 0.0042$ Daily Rank IC, $+1.77 \pm 0.34$ annualized net Sharpe (DSR = 1.000, paired $t = +13.77$ vs. static baseline), and $+1.89 \pm 0.66$ Sharpe across the 2018/2022 crisis regimes with 0.0% guard leakage.

Ablation Configuration ($M = 5$ seeds, 21 episodes)	Recall@3	Frag. Rate	Pass@3 (%)	Leak Rate	Daily Rank IC	Net Sharpe (Δ vs. Full)
0. Full FIN-SKILLS Closed-Loop System (Ours)	100.0%	0.0%	98.3 $\pm$ 0.7	0.0%	+0.0292 $\pm$ 0.0014	+1.71 $\pm$ 0.14 (0.00)
1. w/o Progressive Routing (Fixed-Chunk BM25 RAG, $k = 5$)	55.0%	25.0%	75.0 $\pm$ 1.9	0.0%	+0.0214 $\pm$ 0.0015	+1.14 $\pm$ 0.13 (−0.57)
2. w/o Point-in-Time Cutoff (Disable <code>as_of</code> temporal gate)	100.0%	0.0%	89.4 $\pm$ 1.4	18.3%	+0.0168 $\pm$ 0.0016	+0.92 $\pm$ 0.15 (−0.79)
3. w/o Counterfactual Guards (CI text warnings + unchecked memory)	100.0%	0.0%	54.7 $\pm$ 2.4	45.3%	+0.0031 $\pm$ 0.0019	+0.08 $\pm$ 0.14 (−1.63)
4. w/o Associative Plasticity (frozen_with_gate control)	100.0%	0.0%	98.3 $\pm$ 0.7	0.0%	−0.0085 $\pm$ 0.0013	−0.58 $\pm$ 0.11 (−2.29)
5. w/o 64-KC Sparse Expansion (Dense linear ordinary_with_gate)	100.0%	0.0%	98.3 $\pm$ 0.7	0.0%	+0.0054 $\pm$ 0.0012	+0.17 $\pm$ 0.09 (−1.54)
6. w/o 2D Panel Balance (Remove <code>check_panel_balance</code> , <code>raw panel</code>)	100.0%	0.0%	98.3 $\pm$ 0.7	0.0%	−0.0419 $\pm$ 0.0018	−0.42 $\pm$ 0.16 (−2.13)

Table 4: Seven-arm component-wise ablation study isolating the impact of progressive skill routing, point-in-time retrieval, counterfactual verification guards, mushroom-body plasticity, sparse Kenyon-Cell expansion, and 2D panel conditioning.

9 Discussion: from execution to useful experience

The empirical study shows that successful tool execution does not guarantee sound investment decisions. Both Qwen models discovered and invoked algorithms without syntax errors, yet neither outperformed unassisted baselines. The failure was not in execution, but in method selection and risk management: 7B concentrated into single assets following momentum signals, while 14B relied on mean reversion during trending phases.

These failure modes point directly to the roles of progressive skill routing, persistent memory, and governed operator evolution. When retrieving from large quantitative repositories, fixed-window chunking severs prose pitfall warnings from executable guard schemas (25.0% $\pm$ 1.5% fragmentation at top-5), whereas two-stage manifest routing preserves atomic skill specifications at 4,039 $\pm$ 42 tokens (−98.91%). Meanwhile, cross-episode mushroom-body memory allows an agent to register that a particular strategy failed under specific volatility conditions, adjusting subsequent selections. Furthermore, replacing unconstrained verbal self-reflection with physically diagnosed Fin-RSI operator mutations eliminates pre-LayerNorm gradient cancellation, restores sequence effective sample size during event bursts (3.62 $\times$), and suppresses small-cap estimation variance via per-channel James–Stein shrinkage and streaming Woodbury covariance adaptation.

Finally, execution checks and cryptographic harness locks prevent the agent from reinforcing spurious strategies. If an agent backtests on unadjusted prices, incorporates future data, or mutates evaluation metrics during self-improvement, it observes artificial profits. Without verification guards and a locked trial ledger, persistent memory and recursive evolution would treat those illusions as positive reinforcement. Verification is

therefore the necessary filter that keeps both experiential learning and algorithmic self-improvement grounded in real market dynamics.

10 Conclusion

Access to financial algorithms gives language models the ability to compute, but not the judgment to invest. In this paper, we structured tool use around progressive skill routing, counterfactual execution checks, cross-episode associative memory updates, and governed recursive operator evolution (Fin-RSI). Our experiments demonstrate that while open-weight models readily discover and run specialized algorithms, turning tool execution into effective decision-making requires atomic skill hydration, point-in-time verification guards, feedback-driven synaptic plasticity, and Pareto-governed self-improvement.

Limitations

This study has three primary operational scopes that bound its interpretation. First, our backtest evaluations and autonomous portfolio decisions execute on daily currency reference rates and end-of-day equity bars with parametric transaction-cost curves (5–10 bps) rather than full Level-3 limit order books; live intraday institutional execution introduces stochastic queue-priority decay, partial fills, and dynamic short-borrow fee spikes during crowded liquidations. Second, while our open-weight Beacon audit matrix rigorously characterizes 3B–14B coding models across $M = 5$ seeds, scaling multi-round counterfactual repair beyond five dialogue turns can exhaust context budgets on smaller backbones unless paired with automated AST patch localization. Third, while `check_panel_balance` effectively enforces two-dimensional company $\times$ year cell quotas ($q_{c,y} \leq 36$) across 207,742 daily multi-source records, extending real-time stratified reservoir sampling to sub-second global tick streams requires distributed state coordination across market

ingestion partitions.

Ethics and Reproducibility

All models, tools, and evaluation benchmarks adhere to open, reproducible protocols. Software environments, execution digests, and audit traces are publicly available to facilitate independent verification.

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A Supporting audit and validation studies

This appendix details the execution and feedback validation checks, defect recall benchmarks, and empirical audits across model tiers.

A.1 Independent evaluation environment

We evaluate agent portfolio decisions in a synthetic market environment with full ground-truth transparency, providing quoted prices, corporate split actions, delisting events, and fundamental disclosures. A submitted strategy script returns portfolio weights, alongside a written report stating its expected Sharpe ratio. Session- t portfolio weights must use only information available by the close of session $t - 1$.

An independent oracle evaluator verifies submissions by perturbing future data, checking for post-delisting exposure, and recalculating performance under fixed transaction costs. Oracle grades are generated after the submission receipt is saved and are never exposed to the agent during its decision turns.

This oracle separates genuine investment skill from data leakage. In this environment, an honest strategy trading strictly lagged features achieves an empirical ceiling Sharpe of 1.63. In contrast, look-ahead joins (e.g., using current-day closing prices before the close) produce inflated paper Sharpe ratios up to 71.5. Abnormally high back-test performance thus serves as an unambiguous signature of look-ahead leakage.

A.2 Validation benchmarks

Defect detection benchmark. We evaluate 13 core research guards against 12 planted quantitative defect families, including look-ahead feature joins, unadjusted stock split prices, future earnings disclosure leaks, and survivorship bias. Across three seeded synthetic market worlds, the guards achieve 100% recall (36/36 planted defects caught) with zero false alarms on clean reference code.

Reporting discrepancy and compliance paradox. Table 5 measures reporting discrepancy: the absolute difference between the Sharpe ratio claimed by the agent in its written report and the ground-truth Sharpe ratio computed by the oracle evaluator.

Without tools, Opus exhibited a reporting gap of 0.188, which decreased to 0.034 when given

Model and condition	Mean absolute Sharpe gap
Opus, no library	0.188
Opus, with library	0.034
Haiku, no library	0.574
Haiku, with library	11.949

Table 5: Reporting discrepancy between agent-claimed Sharpe ratios and ground-truth oracle evaluations.

execution guards (81.9% reduction). In contrast, Haiku displayed a compliance paradox: when provided with the library, it frequently wrote in its text rationale that it had verified its strategy with risk guards, but never actually invoked the guards in its code execution trace. This superficial compliance inflated its discrepancy to 11.949, demonstrating that smaller models require automated code execution verification rather than relying on self-reported compliance.

Social sentiment and multi-source corpus verification. Across our consolidated 4,233,499-record (4.23M, 1.25 GB) cross-market financial corpus (`shards_v4`: 2,272,556 EastMoney Guba posts across 464 tickers, 1,544,478 A-share corporate disclosures across 5,122 tickers, 281,981 Xueqiu longitudinal posts across 1,004 tickers, and 134,484 US StockTwits messages across 81 tickers) and 2,980 verified KOL profiles (1,445 CN, 1,535 US; 35,772 timestamped predictions over 799 trading dates), naive follower-weighted sentiment exhibited negative out-of-sample rank correlation with subsequent asset returns (Rank IC = -0.0115): blindly following raw social volume lost money. By auditing each account’s strictly lagged prediction accuracy and applying point-in-time Bayesian credibility gating (including sign-flipping persistently anti-predictive accounts), out-of-sample Daily Rank IC flipped to $+0.0273$ ($\Delta = +0.0388$, block-bootstrap 95% CI $[+0.0295, +0.0480]$) and reached $+0.0327$ under 60-day rolling adaptive recalibration. This confirms that unstructured text signals require empirical track-record verification before driving investment decisions.

A.3 Four-condition audit feasibility

We evaluated models under four audit conditions: unassisted baseline (C0), text guidance only (C1), optional executable guards (C2), and mandatory automated audit (C3).

Operational findings. Table 6 presents results for Qwen2.5-Coder 14B under automated audit constraints. The acceptance rate was 67% under C0 and C1, 100% under C2 (optional guards), and 0% under C3 (mandatory audit), where runs reached the turn limit while attempting to fix all detected code defects. Mean token consumption under C3 was 47,491 compared to 13,877 under C2, showing that automated verification compels the agent to engage in substantially deeper code debugging before finalizing allocations.

A reference quantitative script passed all verification and timing checks on the market benchmark, confirming the end-to-end viability of the automated verification pipeline. Resolving the four harness bottlenecks in Task Interface v2 (BEACON_POSTFIX_RESULTS.json and FOLLOWUP_V2_BUNDLE_MANIFEST.json: explicit "date" index headers across all matrix CSVs, integer-rounded volume.csv perturbation in same_session_probe, "date" column promotion in oracle.py, and bfloat16 sampling) eliminated all ungradable outputs (ungradable_accepted = 0 across all 36 post-fix cells) and demonstrated that mandatory audit (C3) achieves 0.000 incorrect_accepted_per_attempt and 100.0% (8/8) correct_among_accepted.

A.4 Two-dimensional company×year panel balance (check_panel_balance)

Raw financial text crawls across our 4,233,499-record (1.25 GB) multi-source repository (2,272,556 eastmoney_guba, 1,544,478 announcements, 281,981 unified_longitudinal, and 134,484 stocktwits) exhibit severe two-dimensional concentration: mega-cap tickers dominate cross-sectional volume (Company Gini = 0.466–0.540, Max/Median = 20.2×) while recent years (2024–2026) overshadow earlier market regimes (2018–2021). Our 37th research guard, check_panel_balance (guards/panel_balance.py), audits both (Company Gini, Year Gini) and enforces a stratified cell quota $q_{c,y} = \min(n_{c,y}, Q)$ ($Q = 36$). Table 7 evaluates all four panel-conditioning policies across 207,742 aligned multi-source rows (107,999 return observations, 2018–2026, 7 market boards: SH600, SH601_603, SH688, SZ000, SZ002, SZ300, US_Equities). Combining two-dimensional

company×year balancing with point-in-time credibility gating (Cond. D) lifts Mean Daily Rank IC from -0.0416 ± 0.0018 to $+0.0318 \pm 0.0014$ ($\Delta\text{Rank IC} = +0.0734$), reduces Cross-Year IC dispersion by 46.9% ($0.01532 \rightarrow 0.00844$) and Cross-Board dispersion by 53.8% ($0.02973 \rightarrow 0.01304$), and improves net annualized Sharpe by $\Delta\text{Sharpe} = +3.17$ from -1.99 ± 0.08 to $+1.18 \pm 0.06$ ($t = 20.14, p = 1.2 \times 10^{-11}$, Max DD = -10.82%).

A.5 JEV System-One calibration and MMAN multimodal CPCV audit

Finally, Table 8 summarizes the out-of-sample evaluation of (i) the **JEV System-One (RLCD) Isotonic Calibration & 80th-Percentile Volatility Gate** (93,782 OOS test observations across 2021–2025) against uncalibrated raw JEV and pure point-in-time hard-threshold gating, and (ii) the **Multi-Modal Attention Network (MMAN)** 5-fold Purged Combinatorial Cross-Validation (5-day purge window, 10-day embargo, mean CPCV Daily Rank IC = $+0.0275$, positive-IC day ratio = 57.5%) and 2023–2025 out-of-sample holdout across 727 trading days. Isotonic calibration with learned Platt temperature $T = 1.079$ reduces 10-bin Expected Calibration Error (ECE) by 76.4% ($0.0408 \rightarrow 0.0096$) and Brier score from 0.2524 to 0.2485, lifting 5-year all-weather annualized Sharpe from $+1.392$ (Regime C hard-threshold PIT gate) to $+1.986$ ($\Delta\text{Sharpe} = +0.594$, CAGR = $+12.44\%$, Max DD = -3.14% , Calmar = 3.96, monthly win rate = 82.8%). Meanwhile, replacing sparse uncalibrated social posts (Rank IC = $+0.0131$) with fact-rewritten structured catalysts and a Barra-orthogonal dual head lifts 5-day Daily Rank IC to $+0.0508$ ($t = 11.50, p < 10^{-29}$) and excess Sharpe IR to $+0.958$ while compressing maximum Barra style-exposure correlation from 0.4357 to 0.0208.

A.6 Governed financial recursive self-improvement (Fin-RSI) and three-way Pareto audit

To evaluate whether a financial research agent can autonomously improve its underlying multimodal pooling, shrinkage, and covariance-adaptation operators without falling into backtest overfitting or reward hacking, we audit four evolutionary generations across $M = 5$ disjoint random seeds (20260923–20260927) on the full 207,742-row

Model	Condition	Accepted/Planned	Mean tokens	Primary status
14B	C0: unassisted	2/3	13,535	1 × turn limit
	C1: text guidance only	2/3	13,095	1 × turn limit
	C2: optional guards	3/3	13,877	all accepted
	C3: mandatory audit	0/3	47,491	3 × turn limit
7B	all (12 cells)	0/12	—	turn limit / syntax error
32B	all (12 cells)	0/12	0	environment error

Table 6: Audit feasibility results across open-weight models under automated verification constraints.

Panel conditioning policy (207,742 rows from 4.23M corpus, $M = 5$)	Daily Rank IC	Year IC Std	Board IC Std	Net Sharpe	Δ Sharpe (t -stat)	Max DD
A: Unbalanced raw panel (Gini _C = 0.490, Gini _Y = 0.433)	-0.0416 ± 0.0018	0.01532	0.02973	-1.99 ± 0.08	Baseline	-80.83%
B: Company-balanced only ($q_c \leq Q_c$, Gini _C = 0.280)	+0.0051 ± 0.0011	0.01064	0.00906	-0.16 ± 0.05	+1.84	-18.79%
C: Year-balanced only ($q_y \leq Q_y$, Gini _Y = 0.134)	+0.0086 ± 0.0012	0.00896	0.01320	+0.18 ± 0.05	+2.18	-26.40%
D: 2D Company × Year balanced + PIT (check_panel_balance)	+0.0318 ± 0.0014	0.00844 (↓ 46.9%)	0.01304 (↓ 53.8%)	+1.18 ± 0.06	+3.17 ($t = 20.14$)	-10.82%

Table 7: Two-dimensional company × year panel balance ablation ($M = 5$ seeds: 11, 23, 37, 42, 73) across 2018–2026 and 7 equity boards (Δ Rank IC = +0.0734, Δ Sharpe = +3.17, $t = 20.14$, $p = 1.2 \times 10^{-11}$).

two-dimensional company × year balanced panel ($n = 107,999/207,742$ evaluated out-of-sample return observations across 7 equity boards, 2018–2026). Prior to each evolutionary epoch, the sandbox verifier validates the SHA-256 cryptographic signature of the read-only evaluation harness (HARNESS_LOCK.json), enforces all 37 executable counterfactual guards (check_panel_balance, check_safe_asof, check_pit_fundamentals, and check_trial_ledger), and executes three physical diagnostic probes on the multimodal feature stream:

1. Pre-LayerNorm Scale-Cancellation

Probe: Direct scalar salience weighting prior to LayerNorm ($\hat{\mathbf{h}}_t = w_t \mathbf{h}_t$) suffers severe gradient cancellation ($\|\partial \mathbf{y} / \partial w_t\|_2 = 4.8 \times 10^{-5}$), whereas relocating salience pooling into post-encoder value space restores effective gradient norm to 1.74.

2. Intraday Burst Sequence ESS Probe:

Unbounded exponential attention weights collapse onto single outlier posts during earnings and forum bursts (ESS = 4.61), whereas temperature-bounded tanh activation ($\tau = 2.5$) paired with Log-Sum-Exp intraday burst partition conservation lifts effective sample size to 14.72 (3.62 × improvement).

3. Subspace SNR & Tail-Slice Divergence

Probe: Uniform scalar shrinkage degrades out-of-sample Rank IC on sparse small-cap and STAR/ChiNext tickers (+0.0211),

whereas per-channel subspace precision gating with positive-part James–Stein shrinkage (James and Stein, 1961) lifts sparse-slice Rank IC to +0.0290.

Table 9 reports the complete multi-seed Pareto ledger anchored against the full-budget dense multimodal reference first, followed by unconstrained verbal self-reflection (Unconstrained Verbal RSI), static production baselines (Static MMAN and Static MMAN + JEV-64KC Baseline), and the three governed Fin-RSI generations. While unconstrained verbal self-reflection without a locked evaluation harness or physical probes permits 38.5% silent look-ahead/overfitting violations and suffers severe out-of-sample drawdown degradation (Net Sharpe = +0.18 ± 0.23, Max DD = -22.47%), each governed Fin-RSI generation achieves strict 0.0% leakage and monotonically improves macro alpha, sparse small-cap Rank IC, and 2018/2022 crisis-regime Sharpe. The promoted **Fin-RSI Gen-3 Champion** (combining post-encoder bounded-salience pooling, per-channel James–Stein shrinkage, streaming rank-1 Sherman–Morrison–Woodbury inverse-covariance adaptation, and Bernoulli Fisher information gating $g_{\text{Fisher}}(p_t) = 4p_t(1 - p_t)$) achieves an out-of-sample Daily Rank IC of +0.0363 ± 0.0042 (IC IR = +3.46 ± 0.51), lifts sparse small-cap/STAR slice Rank IC to +0.0347 ± 0.0073 and 2018/2022 crisis Sharpe to +1.89 ± 0.66, and reaches an annualized net Sharpe of +1.77 ± 0.34 (DSR = 1.000, Max DD = -6.84%, paired $t = +13.77$ vs.

Evaluation Suite	Model / Gating Regime (2021–2025 / 2023–2025 OOS)	Primary Signal / Calibration Metric	CAGR / Alpha	Annualized Sharpe / IR	Max Drawdown	Calmar / Style Corr
JEV System-One	Regime A: Uncalibrated Long-Only Baseline	$ECE_{10} = 0.0408$, Brier = 0.2524	+8.60%	+0.491	−18.87%	Calmar = 0.46
	Regime B: Static Market-Neutral (No Gate)	Uncalibrated Raw JEV ($ECE_{10} = 0.0408$)	+7.74%	+1.119	−8.43%	Calmar = 0.92
	Regime C: Pure PIT Hard-Threshold Gate	Uncalibrated + 80% Vol Gate	+9.32%	+1.392	−3.57%	Calmar = 2.61
	Regime D: JEV System-One Isotonic + 80% Vol Gate	$ECE_{10} = 0.0096$ ($\downarrow$ 76.4%), $T = 1.079$	+12.44%	+1.986	−3.14%	Calmar = 3.96
MMAN (727d OOS)	Model 1: Pure Price–Volume BiLSTM Baseline	5d Daily Rank IC = +0.0287 ($t = 8.13$)	+2.17%	+0.202	−36.73%	$ \rho_{\text{Barra}} = 0.3369$
	Model 2: Price–Volume + Sparse Old Social (7-dim)	5d Daily Rank IC = +0.0131 ($t = 4.56$)	+1.53%	+0.149	−32.68%	$ \rho_{\text{Barra}} = 0.4357$
	Model 3: Price–Volume + Fact-Rewritten Catalysts (17-dim)	5d Daily Rank IC = +0.0492 ($t = 14.24$)	+5.80%	+0.628	−48.40%	$ \rho_{\text{Barra}} = 0.1905$
	Model 4: Fact-Rewritten + Barra-Orthogonal Dual Head	5d Daily Rank IC = +0.0508 ($t = 11.50$)	+8.17%	+0.958	−25.99%	$ \rho_{\text{Barra}} = 0.0208$

Table 8: JEV System-One isotonic calibration and 80th-percentile volatility gating (93,782 OOS observations, 2021–2025) alongside MMAN 5-fold CPCV and out-of-sample multimodal evaluation (727 trading days, 2023–2025).

static Gen-0).

Evolutionary Arm ($M = 5$ seeds, $n = 107,999/207,742$)	Governance & Operator Mechanism	OOS Daily Rank IC	Annualized IC IR	Sparse Ticker IC	2018/2022 Crisis Sharpe	OOS Net Sharpe (DSR)	Max DD / ESS Ratio
Full Dense Multimodal Reference (100% Budget)	Uncompressed 16D Multimodal Anchor	$+0.0315 \pm 0.0042$	$+3.04 \pm 0.44$	$+0.0264 \pm 0.0081$	$+1.59 \pm 0.52$	$+1.41 \pm 0.40$ (1.000)	$-8.85\% / 3.10\times$
Unconstrained Verbal RSI (w/o Harness Lock)	Open Prompt Self-Critique (38.5% Leak)	$+0.0045 \pm 0.0024$	$+0.42 \pm 0.22$	$+0.0029 \pm 0.0044$	-0.34 ± 0.97	$+0.18 \pm 0.23$ (0.000)	$-22.47\% / 1.00\times$
Static MMAN Baseline	Barra-Orthogonal Dual-Head Network	$+0.0281 \pm 0.0036$	$+2.67 \pm 0.40$	$+0.0214 \pm 0.0057$	$+1.07 \pm 0.91$	$+1.26 \pm 0.36$ (1.000)	$-11.81\% / 1.18\times$
Static MMAN + JEV-64KC Baseline (Gen-0)	Isotonic 80% Vol Gate + 64-KC Memory	$+0.0283 \pm 0.0038$	$+2.73 \pm 0.40$	$+0.0236 \pm 0.0069$	$+1.10 \pm 0.61$	$+1.16 \pm 0.22$ (1.000)	$-10.53\% / 1.34\times$
Fin-RSI Gen-1 (Bounded-Saliency Pooling)	Value-Space $\exp(\tau \tanh(z/\tau))$ + LSE Burst	$+0.0287 \pm 0.0041$	$+2.75 \pm 0.47$	$+0.0211 \pm 0.0075$	$+1.32 \pm 0.55$	$+1.30 \pm 0.13$ (1.000)	$-8.19\% / \mathbf{3.48\times}$
Fin-RSI Gen-2 (+ Subspace Stein Denoising)	Per-Channel α_d Gate + Positive-Part JS	$+0.0315 \pm 0.0041$	$+3.01 \pm 0.47$	$+0.0290 \pm 0.0075$	$+0.77 \pm 0.34$	$+1.51 \pm 0.28$ (1.000)	$-6.79\% / \mathbf{3.54\times}$
Fin-RSI Gen-3 Champion (Promoted)	+ Streaming Rank-1 Woodbury & Fisher Gate	$+0.0363 \pm 0.0042$	$+3.46 \pm 0.51$	$+0.0347 \pm 0.0073$	$+1.89 \pm 0.66$	$+1.77 \pm 0.34$ (1.000)	$-6.84\% / 3.62\times$

Table 9: Four-generation Governed Financial Recursive Self-Improvement (Fin-RSI) Pareto audit across $M = 5$ disjoint seeds (20260923–20260927) on the 207,742-row 2D company $\times$ year balanced panel ($n = 107,999/207,742$ evaluated OOS return observations, 2018–2026). All governed Fin-RSI generations enforce SHA-256 harness locking and pass all 37 counterfactual verification guards (0.0% leakage).

Component	Entries
Skill documents	129
Algorithm-backend catalog	48
Registered research guards	37
JSON-callable tools	63

Table 10: Inventory of quantitative research library interfaces.

B Library implementation and interface scope

Knowledge and executable interfaces. Domain skills route tasks to relevant libraries, while per-library skills provide conventions, input schemas, and common failure modes. The Python package exposes catalog search, skill loading, and reference retrieval. Packaged skill documents and scripts are compiled from validated repository sources, with all skill entries documenting primary-source verifications and compatibility metadata. Table 10 summarizes the core library components.

Algorithm selection and execution. The algorithm catalog specifies input requirements, dependencies, and supported tasks. Implemented families include portfolio construction, forecasting, volatility and risk estimation, technical signals, and execution schedules. Recommendation rules filter candidates based on task specifications and input data constraints. The execution layer avoids silent fallbacks, raising explicit errors when dependencies or data requirements are not met.

Reusable models and lifecycle. The library supports programmatic Python execution as well as agentic tool dispatch. Forecasting, regression, and classification adapters expose a uniform fit-predict lifecycle, supporting ARIMA, Ridge, random forests, LightGBM, and XGBoost. Fitted models and estimators can be serialized and restored alongside environment metadata. The model registry additionally provides an adapter for the bio-inspired memory architecture via `create_model("fly_memory")`, supporting prediction, verified feedback updating, and state serialization.

Built-in retrieval and model composition. The RAG interface chunks packaged skills or external documents and retrieves passages using BM25 and neural rerankers. Passages retain document identifiers, source offsets, and availability timestamps;

point-in-time retrieval excludes documents unavailable at the requested decision cutoff. Citation checks validate references against retrieved passage identifiers. The pipeline freezes candidates, temporal cutoffs, and context budgets prior to neural reranking. On the temporal retrieval benchmark (RAG_JEV_RESULTS.json), point-in-time `as_of` filtering records 0 future-document leaks while lifting `Recall@3` / `nDCG@3` from 50.0% / 35.5% (fixed skill headers) to 100.0% / 100.0% under two-stage BM25 + semantic reranking. Across all 129 skills (2052 chunks) on 60 evaluation tasks, two-stage progressive manifest routing achieves 100.0% skill recall with 0.0% code-vs.-guidance fragmentation and 98.3% `Pass@3` at 4,039 mean tokens (−98.9% vs. full-library concatenation), compared to 55.0% recall and 25.0% fragmentation (75.0% `Pass@3`) for fixed-chunk BM25 (`top_k=5`).

Information availability and point-in-time processing. News and disclosure collection supports RSS/Atom feeds, static pages, GDELT, and official regulatory sources. Collection records distinguish publication time from observation time and retain revision histories. Point-in-time digests filter future, stale, and undated records while deduplicating canonical URLs.

Model choice and rule-based advice. An agent can inspect skills, discover candidate algorithms, execute chosen methods with custom parameters, and incorporate numerical outputs into its decision logic. Market-state categorization provides heuristic regime indicators (trend, range, stress) that agents can inspect as additional evidence alongside price tables.

Research workflow and audit boundaries. Research interfaces enforce chronological validation splits, out-of-sample holdout reporting, and standardized trial records. Python and JSON tool interfaces return structured dictionary outputs, ensuring that all intermediate calculations can be logged and audited.

Local decision and reranking adapters. To support reproducible offline auditing without hosted API credentials, FIN-SKILLS provides unified local checkpoint adapters spanning point-wise financial encoders (ProsusAI/finbert, BAAI/bge-reranker-v2-m3, and our JEV System-One calibrated router) alongside ex-

ternal negative-control adapters for Kev¹ and Laya². All adapters log exact checkpoint hashes, input token budgets, and structured tool receipts, while distinguishing true isotonic probability calibration (Table 8) from uncalibrated heuristic probability normalization.

Three-tier finance-native model matrix vs. non-financial negative controls. A recurring failure mode in financial agent evaluations is conflating general-purpose routing toys or brittle text parsers with domain-calibrated financial architectures. To isolate model-level and parser-level failure modes, we organize our evaluation around a **Three-Tier Finance-Native Model Matrix** contrasted against **Off-the-Shelf Non-Financial and Uncalibrated Negative Controls**:

- **Negative Controls (Off-the-Shelf Non-Financial Rerankers & Naive Parsers):** We include autoregressive listwise rerankers (jaredpalmer/kev-0.8B and kev-4B), short-context classifiers (NandhaKishorM/laya, 128-token option budget), and unconstrained general/code checkpoints (Mistral-Nemo-2407 with regex tokenizer instability and Qwen2.5-Coder-14B coupled to a naive json.loads final-string parser) strictly as negative controls to expose position bias, context truncation, and wrapper-induced parse collapse.
- **Tier 1 (Financial Semantic Routing, Filing Reranking & Schema-Guided Receipt Verification):** For skill routing and SEC filing evidence retrieval, FIN-SKILLS combines fast vectorized lexical search (BM25S) with pointwise financial domain encoders (FinBERT, ProsusAI/finbert; and BGE-Reranker-v2-m3, Chen et al. 2024), our **JEV System-One Calibrated Two-Stage Router**, and **Schema-Guided Calculator Receipt Verification** (Fin-R1 / DeepSeek-R1-Distill program synthesis + RAGPipeline citation gate).
- **Tier 2 (Financial Specialist Agent & Classical Quantitative Baselines):** In our end-to-end backtest benchmark (Tables 3–4), we compare directly against domain-specific

financial agent frameworks—FinMem (Yu et al., 2024), TradingAgents (Xiao et al., 2024), and FinAgent (Zhang et al., 2024)—as well as classical quantitative allocation baselines—HRP (López de Prado, 2016) and TSMOM (Moskowitz et al., 2012)—achieving 96.2% Pass@1, 0.0% survivorship/look-ahead leakage, and +1.71 annualized net Sharpe.

- **Tier 3 (Deep Multimodal Network MMAN, JEV System-One & Governed Fin-RSI on the 4.23M Corpus):** At the cross-market production scale (4,233,499 timestamped records across 5,122 A-share and 81 US equities; Tables 7–9), enforcing 2D (Company × Year) panel balance (207,742 rows, Year Gini < 0.04, paired $t = 20.14$) enables our MMAN Barra-orthogonal dual-head network to reach out-of-sample Daily Rank IC = +0.0508 ($t = 11.50$), our static JEV System-One isotonic-calibrated decision gate to achieve a 5-year OOS annualized net Sharpe of +1.986 (Max DD = −3.14%), and our Governed Financial Recursive Self-Improvement (Fin-RSI) champion operator to achieve $+1.77 \pm 0.34$ annualized net Sharpe (Daily Rank IC = $+0.0363 \pm 0.0042$) under zero-leakage sandbox governance.

Routing permutation invariance and two-stage SEC filing retrieval. Part A of Table 11 evaluates 108 skill-routing queries (36 skills × 3 query styles) under a seeded shuffled candidate order and its exact reversal. Because Kev-0.8B and Kev-4B score all candidates jointly inside an autoregressive prompt, reversing candidate order flips their Top-1 decision on 36/108 (33.3%) and 24/108 (22.2%) of queries, respectively, while Laya rejects all 108/108 queries prior to inference because full skill specifications exceed its 128-token per-option budget. In contrast, pointwise scoring with BM25S (76/108), FinBERT (82/108), and BGE-Reranker-v2-m3 (87/108) guarantees exact permutation invariance (0/108 flips), and our JEV System-One calibrated two-stage router lifts Top-1 accuracy to 94/108 (87.0%) and Recall@3 to 104/108 (96.3%) with zero order flips. To test whether two-stage reranking transfers to long-form corporate filings, we also

¹<https://github.com/jaredpalmer/kev>

²<https://github.com/NandhaKishorM/laya>

Table 11: Tier-1 finance-native semantic routing (108 skill-routing queries across 36 skills $\times$ 3 phrasings) and multi-step SEC 10-K numerical reasoning (32 FinQA questions across 16 companies; Chen et al. 2021) versus off-the-shelf non-financial and uncalibrated negative controls. Pointwise financial encoders and the JEV System-One calibrated router eliminate the 22.2%–33.3% candidate-order reversal flips of autoregressive listwise Kev and the 100% context-truncation rejections of Laya. On FinQA, replacing naive `json.loads` string parsing (which suffered 30–32/32 Markdown fence and turn-limit exceptions despite 30/32 = 93.8% valid intermediate calculator calls) with schema-guided receipt verification and our two-stage financial reranker lifts verified execution accuracy from 0–6.2% to 87.5%–93.8% with 0/32 parse crashes and 0/32 unverified hallucinations.

Evaluation Task & Tier	Model / Routing & Verification Pipeline	Shuf. Top-1	Rev. Top-1	Recall@3 / Evid.	Order Flips / Err.	Verified Acc.
<i>Part A: Skill Catalog Semantic Routing & Order-Reversal Invariance (N = 108 queries, 3 frozen candidates/query)</i>						
Negative Control	NandhaKishorM/laya (128-tok context limit)	0/108	0/108	0/108 (0.0%)	108/108 trunc.	0.0%
Negative Control	jaredpalmer/kev-0.8B (listwise autoreg.)	45/108	50/108	100/108 (92.6%)	36/108 (33.3%)	41.7%
Negative Control	jaredpalmer/kev-4B (listwise autoreg.)	79/108	82/108	100/108 (92.6%)	24/108 (22.2%)	73.1%
Tier-1 Lexical	BM25S Pointwise Lexical Baseline	76/108	76/108	100/108 (92.6%)	0/108 (0.0%)	70.4%
Tier-1 Finance	FinBERT (ProsusAI/finbert pointwise)	82/108	82/108	100/108 (92.6%)	0/108 (0.0%)	75.9%
Tier-1 Cross-Enc	BGE-Reranker-v2-m3 (pointwise cross-enc.)	87/108	87/108	100/108 (92.6%)	0/108 (0.0%)	80.6%
Tier-1 Ours	JEV System-One Calibrated Two-Stage Router	94/108	94/108	104/108 (96.3%)	0/108 (0.0%)	87.0%
<i>Part B: FinQA Multi-Step SEC 10-K Numerical Reasoning & Receipt Verification (N = 32 questions across 16 companies)</i>						
Negative Control	Mistral-Nemo-2407 + Kev-4B/BGE + Naive <code>json.loads</code>	—	—	0/32 (0.0%)	32/32 (100%)	0/32 (0.0%)
Negative Control	Qwen2.5-Coder-14B + Kev-4B + Naive <code>json.loads</code>	—	—	11/32 (34.4%)	32/32 (100%)	0/32 (0.0%)
Negative Control	Qwen2.5-Coder-14B + BGE + Naive <code>json.loads</code>	—	—	20/32 (62.5%)	30/32 (93.8%)	2/32 (6.2%)
Tier-1 Lexical	BM25S Retrieval + Calculator Receipt Gate	—	—	21/32 (65.6%)	0/32 (0.0%)	21/32 (65.6%)
Tier-1 Finance	FinBERT + BGE-v2-m3 + Calculator Receipt Gate	—	—	25/32 (78.1%)	0/32 (0.0%)	27/32 (84.4%)
Tier-1 Ours	Qwen2.5-Coder-14B + Schema-Guided Receipt Recovery	—	—	28/32 (87.5%)	0/32 (0.0%)	28/32 (87.5%)
Tier-1 Ours	JEV Two-Stage Reranker + Fin-R1 Calculator Gate	—	—	30/32 (93.8%)	0/32 (0.0%)	30/32 (93.8%)

indexed 53,686 nonempty pages across 366 local SEC PDFs for the 150-question FinanceBench benchmark (Islam et al., 2023). Vectorizing lexical retrieval with BM25S reduces median query latency from 5.504 s to 0.0044 s (0.125 s index build), while adding second-stage BGE cross-encoder reranking over the top-25 lexical pages improves Top-1 evidence-page hit rate by +55.6% (9/150 $\rightarrow$ 14/150) and Top-5 hit rate by +53.3% (15/150 $\rightarrow$ 23/150).

FinQA receipt verification, citation gating, and flat tool-catalog guidance. Part B of Table 11 resolves an instructive evaluation pathology on 32 FinQA numerical reasoning questions (Chen et al., 2021). Under the legacy unconstrained harness, pairing Mistral-Nemo-2407 or Qwen2.5-Coder-14B with a naive `json.loads` parser produced 0/32 and 2/32 correct completions. Trajectory forensics revealed that in 30/32 (93.8%) of runs, the agent *did* execute valid intermediate calculator tool calls, but naive final-string parsing threw `JSONDecodeError` whenever the model wrapped its final JSON in Markdown ````json` fences (25/32 runs on Qwen; 7/32 on Mistral) or hit the 8-turn LlamaIndex limit (23/32 on Mistral). Replacing naive string parsing with **Schema-Guided Receipt Recovery** recovers 28/32 (87.5%) verified answers on Qwen2.5-Coder-14B with 0/32 parse errors, while our **JEV Two-Stage Reranker + Fin-R1 Calculator Gate** achieves 30/32 (93.8%) gold-

evidence recall and verified execution accuracy. Simultaneously, enforcing the RAGPipeline **Citation Gate**—which rejects any numerical claim lacking a verified calculator receipt and retrieved chunk ID—reduces Incorrect Accepted FinQA answers from 5/32 down to **0/32**. Across a 15-tool LlamaIndex catalog ablation comparing Generic (one-line descriptions), Organized (nested category headers), and Flat guidance (concise inline operational rules), **Flat guidance** achieves the highest verified completion (28/32 = 87.5% on Qwen with 0 format errors and 0 call-limit exceptions, vs. 26/32 Organized and 25/32 Generic; and 13/32 vs. 3/32 on Mistral), demonstrating that flat, decision-oriented inline rules outperform deep tool hierarchies.

Storage crash recovery and arrival-order vs. point-in-time revision semantics. Finally, we audited the SQLite collection and monitoring state store across four process-termination boundaries (before_save, mid_batch, pre_commit, and post_commit, three repeats each = 12/12 SIGKILL injections) and scale runs of 100, 1,000, and 10,000 synthetic events (0.154 s / 0.404 s / 3.036 s ingest; 45.8 ms / 56.6 ms / 58.1 ms reopen; 3.89 ms / 4.54 ms / 8.67 ms median latest-query latency). All 12/12 crash tests recovered the exact atomic database state across event records, latest-version pointers, pending alerts, and collection cursors. Crucially, an out-of-order revision injection test confirmed

a vital architectural boundary: the storage engine's operational `latest` pointer tracks *arrival order* rather than chronological publication order, so a delayed restatement or backfilled historical revision arriving last becomes the current operational record. Consequently, historical backtests must never treat storage arrival pointers as point-in-time knowledge, motivating our mandatory `check_pit_fundamentals` and `check_safe_asof` runtime guards.